

Chapter 1, Problem 4P

Problem

A total charge of 300 C flows past a given cross section of a conductor in 30 seconds. What is the value of the current?

Step-by-step solution

Step 1 of 2

Write the expression for current flowing through conductor.

$$i = \frac{q}{t}$$

Step 2 of 2

The total charge flow (q) through a given cross section of a conductor in 30 seconds is 300 C

. Calculate the value of the current.

$$\begin{aligned} i &= \frac{q}{t} \\ &= \frac{300 \text{ C}}{30 \text{ s}} \\ &= 10 \text{ A} \end{aligned}$$

Therefore, the current through the given conductor is 10 A.